Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/25/2006
PI Name:	Simons, Donald Ph.D.		
Project Title:	Impaired Activation and Signal Transduction in Mouse Splenocytes Cultured in Modeled Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
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Task Description:	Immune inhibition is a well- documented effect of space flight and has been replicated in ground- based experiments in a modeled microgravity (MMG) environment. We hypothesize that immune inhibition seen in space flight is a result of microgravity affecting directly the signaling pathways involved in T cell activation. We propose to examine T cell stimulation in MMG at three different levels of focus, at the global cellular level, at the specific cell level, and at the level of molecular interactions. Our goal is to define the underlying mechanisms responsible for immune inhibition seen in space flight and to develop countermeasures The goal of this study is to specifically identify signal transduction molecules and/or events that are adversely affected by culture in microgravity and lead to the observed inhibition of T cell activation in microgravity culture. Both space flight and ground based data indicate that the cellular immune response of astronauts is inhibited during and immediately following orbital flight. Additional studies have shown that the ability of in vitro cultured T lymphocytes to respond to activating stimulus is dependent on the presence of gravity, and that inhibition is likely due to dysfunctional signal transduction through the T cell receptor. These results indicate		

	that long-term manned spaceflight may require the development of countermeasures to alleviate this inhibition. The development of effective countermeasures will require a detailed knowledge of the specific affects of microgravity on T cell signal transduction. The goal of this research is to provide a map of T cell signal transduction during modeled microgravity culture for the express purpose of future evidence-based countermeasure design.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In vitro culture of T cells in true- and modeled-microgravity provides a unique (non-chemical, epigenetic) means of inhibiting signal transduction; allowing for a minimally invasive examination of the key signaling molecules required for T cell activation. Identification of these molecules in fully functional cells in a severely inhibitory environment will also provide insight into T cell function in less inhibitory situations such as aging. Conversely, identification of key signaling molecules will also provide novel targets for treatment of autoimmune disease.
Task Progress:	The proposed investigation for this project comprised three specific aims: 1.) investigation of the kinetics of inhibition, 2.) investigation of the phenotype and fate of inhibited cells, and 3.) investigation of the signal transduction pathway and gene expression profiles of inhibited cells. The purpose of the first two of these aims is to determine when and where to look, that is at what time during exposure is inhibition specifically due to microgravity induced dysfunction of signal transduction (and conversely, when do the cells become permanently quiescent regardless of the presence or absence of gravity), and where in the T cell subsets does inhibition occur (i.e. is there a difference in CD4 vs. CD8 T cells). When inhibition is occurring and in which cell types it is occurring is vitally important to completion of the third specific aim. This aim seeks to map inhibition in the signal transduction pathway using phospho-specific antibody arrays for key signaling molecules, and to profile the resulting gene expression using cDNA array technology. Investigation of appropriate cell types and time points is extremely important for these techniques due to their high sensitivity. To date, experiments have focused on the first two aims by establishing the period of modeled microgravity (MMG) culture during which signal transduction is specifically inhibited, and by examining the viability and activation status of both T subsets following exposure. These experiments have been carried out by exposing Con A stimulated splenocytes to increasing durations of MMG, and then monitoring their ability to proliferate and express activation markers when returned to 1-g conditions. Our results indicate that a window of time exists during the first 8-hrs of culture in MMG during which inhibition is specifically due to attenuated signaling. Following 8- and 24-hrs of culture, T cells experience a decreased capacity to proliferate in response to mitogen in a manner greater than would be expected if inhibition were due solely to altered signal transduction during exposure. Preliminary results with regard to the second aim of this project have shown that both T subsets are affected identically by MMG exposure. The fate of these cells remains unclear and will require further investigation to complete the second aim. With respect to early events during T cell activation (such as upregulation of the IL-2 receptor high-affinity a-subunit) the cells appear normal after being allowed to recover at 1-g. The long-term viability of these cells during the recovery period, however, was significantly decreased compared to controls following greater than 8-hrs of exposure to MMG. Interestingly, decreased proliferation does not appear to correlate with a loss of T cell viability, indicating that a more profound inhibitory mechanism affecting downstream events may occur during longer exposure times. Current experiments are underway to examine the induction of apoptosis in T cells exposed to MMG for a more detailed investigation of the role of cell death on the loss of proliferation. Investigation of the third specific aim will also proceed during the course of the year. Based on our recent results these experiments will focus on examination of signaling and gene expression during the first 8-hrs of MMG exposure, and will examine the effect of MMG culture on T cells as a single population rather than individual subsets.
Bibliography Type:	Description: (Last Updated: 01/15/2008)
Articles in Peer-reviewed Journals	Simons, D.M.; Gardner, E.M.; Lelkes, P.I. "Dynamic culture in a rotating-wall vessel bioreactor differentially inhibits murine T-lymphocyte activation by mitogenic stimuli upon return to static conditions in a time-dependent manner" J. Appl. Phys. 2005, in press. , Jun-2005